



Basic Principles of Medicine 2
Module: Digestive System and Metabolism | DLA

Peritoneal Relationships

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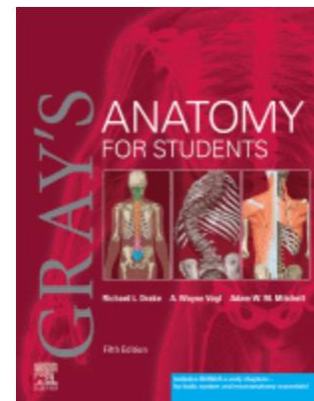
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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Online Textbook: [Chapter 4: Click here](#)
 - Component Parts – Abdominal Cavity
 - Abdominal Wall – Peritoneum
 - Abdominal Viscera
 - Peritoneum
 - Omenta, mesenteries and ligaments
 - Liver
- Pay special attention to the **green (print text)/blue (ebook)** “In the Clinic” boxes for the clinical correlations for this topic

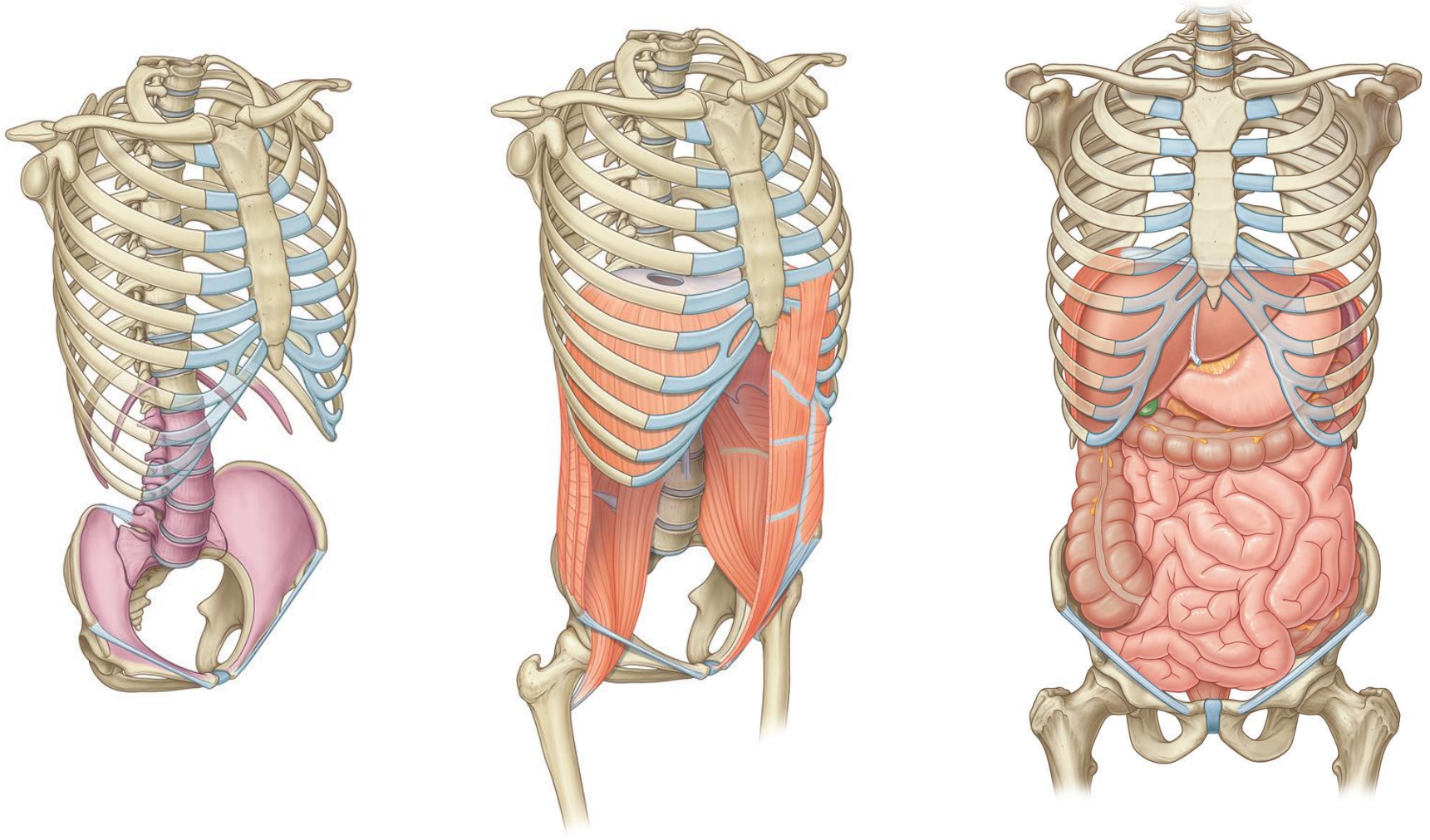


Objectives

- SOM.MK.1.BMP2.3.DM.1.ANAT.1117** Describe the parietal and visceral peritoneum
- SOM.MK.1.BMP2.3.DM.1.ANAT.1118** Describe the extent of the peritoneal cavity, the greater and lesser sacs, and the omental (epiploic) foramen (of Winslow).
- SOM.MK.1.BMP2.3.DM.1.ANAT.1119** Describe the difference between intraperitoneal and retroperitoneal abdominal organs.
- SOM.MK.1.BMP2.3.DM.1.ANAT.1120** Describe the differences between primary and secondary retroperitoneal abdominal organs.
- SOM.MK.1.BMP2.3.DM.1.ANAT.1122** Describe the following peritoneal mesenteries and the abdominal organs they enclose/suspend: "the mesentery", transverse mesocolon and sigmoid mesocolon.
- SOM.MK.1.BMP2.3.DM.1.ANAT.1123** Describe the contents of the supra and infracolic compartments, the location of the paracolic gutters, the vesicouterine pouch, the rectouterine pouch (of Douglas) and the hepatorenal pouch (of Morrison) within the peritoneal cavity.
- SOM.MK.1.BMP2.3.DM.1.ANAT.1125** Describe the following ligaments, their embryologic origin, peritoneal coverings and their function: falciform ligament, round ligament of the liver (ligamentum teres hepatis), coronary, triangular, median, medial and lateral umbilical ligament, greater omentum, gastrocolic, gastrosplenic, splenorenal ligaments, lesser omentum (hepatoduodenal and hepatogastric ligaments) and suspensory ligament of the duodenum (Treitz).

Abdominopelvic cavity

- Abdominopelvic cavity – space bounded by the abdominal walls, diaphragm and pelvis. Contains viscera and peritoneal cavity.



Peritoneum

- Thin, epithelial-like single layer of cells with supporting connective tissue
- Continuous serous membrane which surrounds the organs of abdominal cavity and lines the abdominopelvic wall.
- Secretes small amount of fluid which lubricates the surface of the peritoneum and allows free movement of viscera
- Provides mobility for some & stability for others via reflection & fusion points.

Two layers

- **Parietal peritoneum**
 - Lines internal surface of abdominal wall
- **Visceral peritoneum**
 - Covers and suspends organs

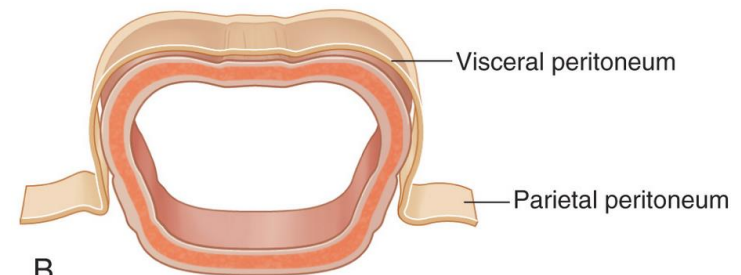
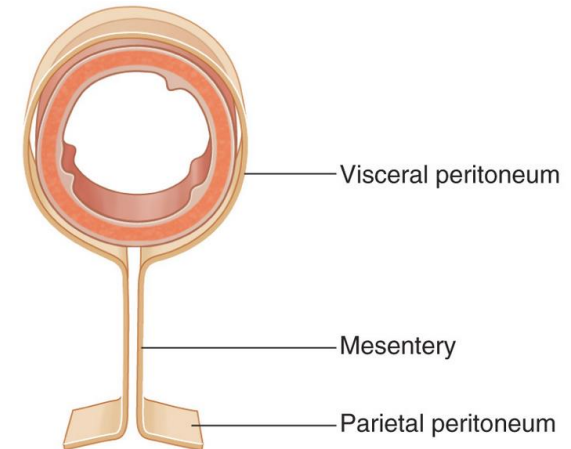
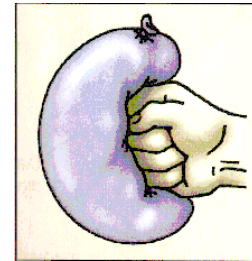
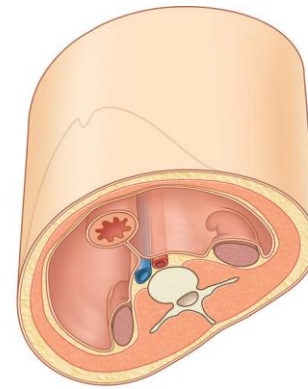


Fig. 4.53

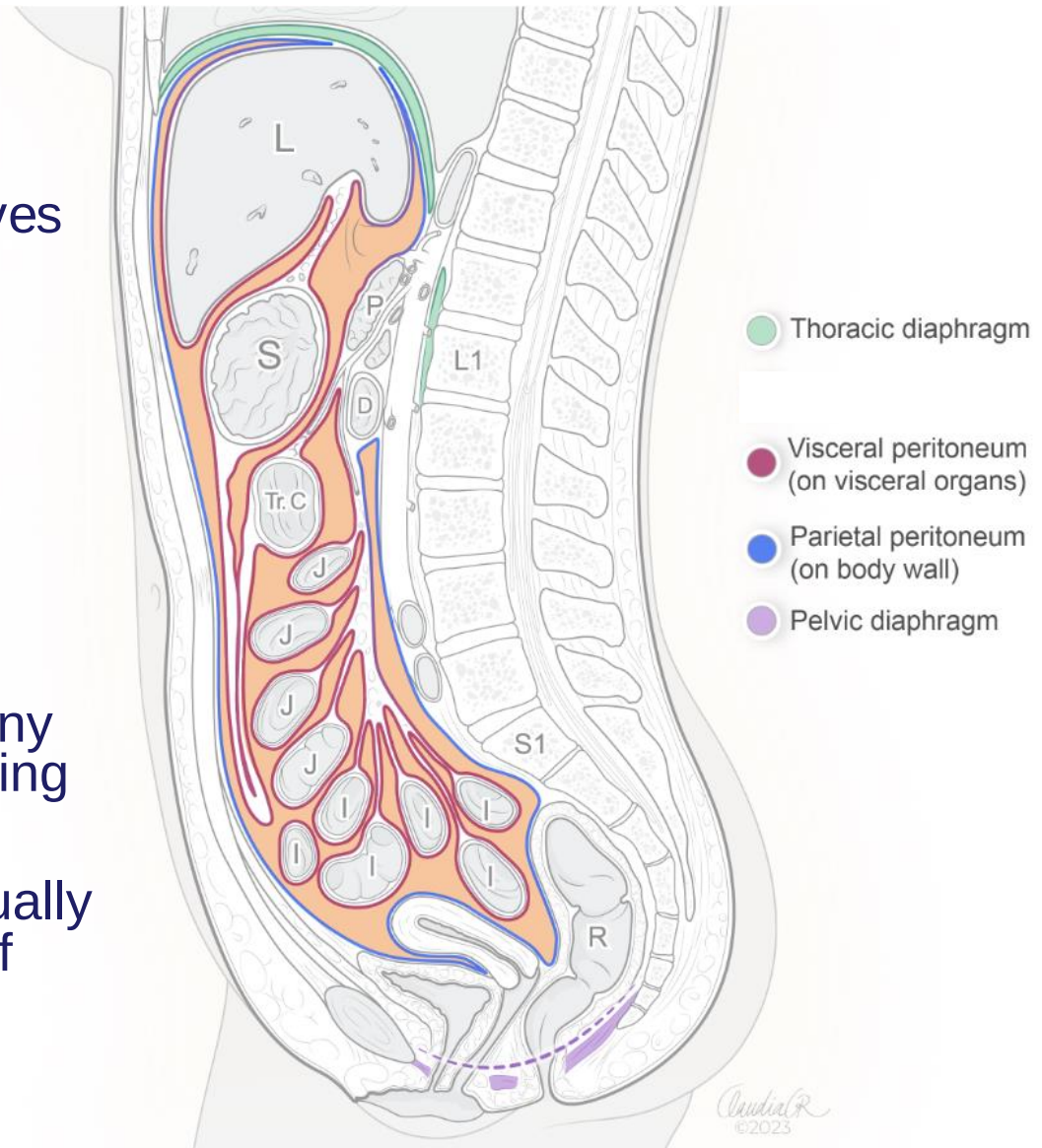
Peritoneal Formation

Parietal peritoneum (somatopleure)

- Innervated by **somatic afferents** from spinal nerves associated with adjacent region of abdominal wall
- Well localized Pain**

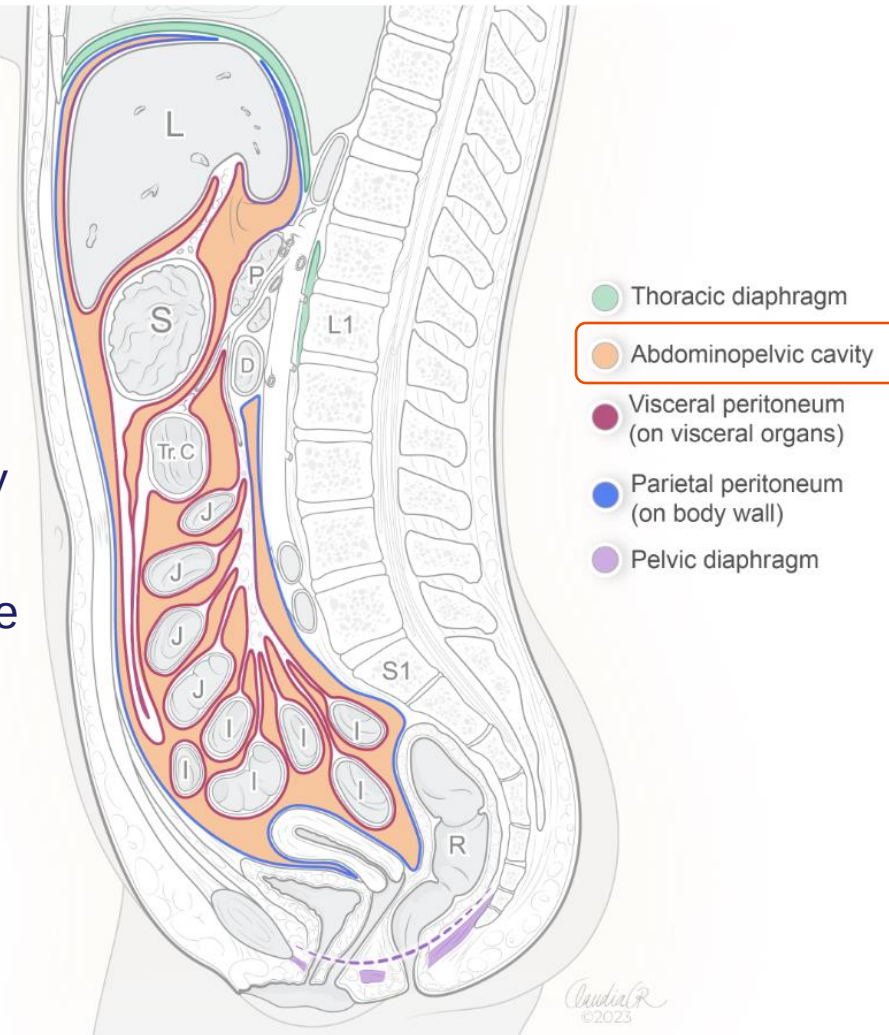
Visceral peritoneum (splanchnopleure)

- Innervated by **visceral afferents** which accompany autonomic nerves supplying organs
- Poorly localized pain**, usually presents as sensations of discomfort

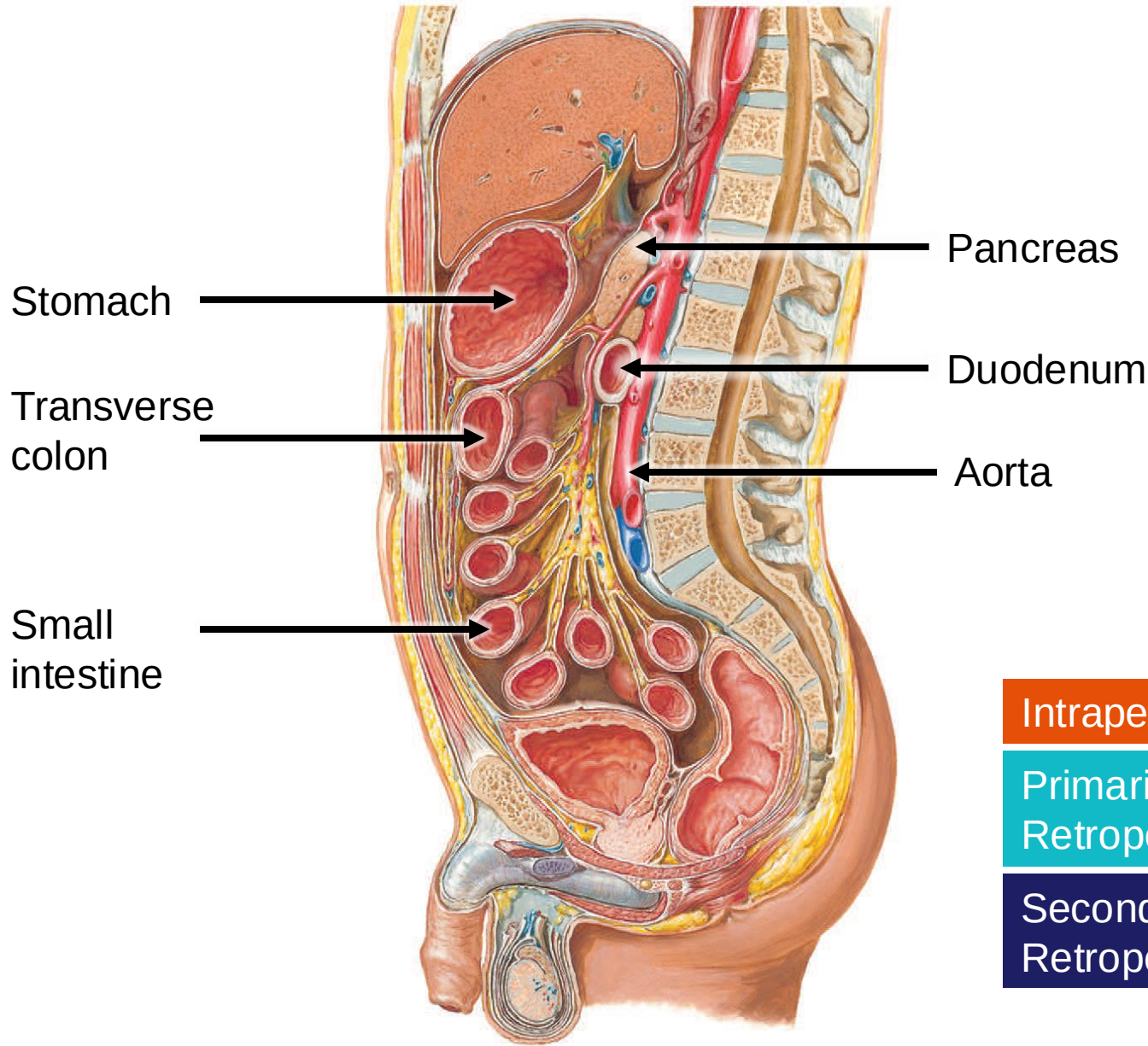


Peritoneal cavity

- *Potential space* between the parietal and visceral layers of peritoneum and **is one continuous space**
- Extends from the diaphragm to the pelvis
 - Male system closed
 - Female open at uterine tube
- **Ascites**
 - Excess fluid in the peritoneal cavity
- **Pneumoperitoneum**
 - Abnormal presence of air/gas in the peritoneal cavity
 - Generally due to a ruptured hollow viscus organ
- **Peritonitis**
 - Inflammation of the peritoneum
- There are no organs in the peritoneal cavity



Intra vs Retroperitoneal



Peritoneal Formation

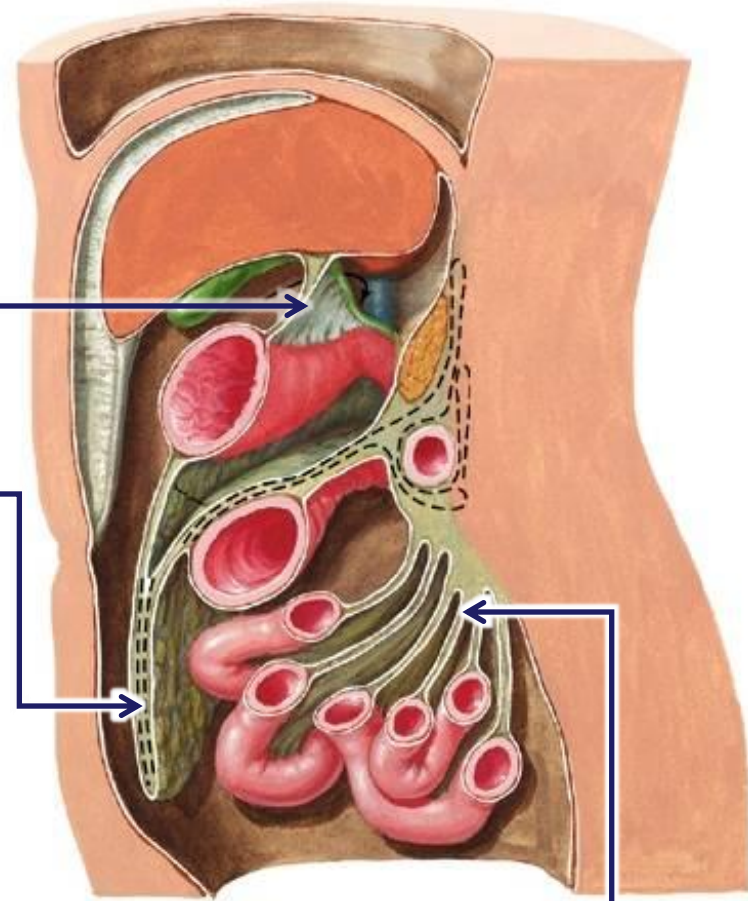
Omentum: double layer of peritoneum passing from the stomach and proximal part of the duodenum to adjacent organs

Lesser omentum: double layer of peritoneum attaching liver to the stomach and first part of duodenum (*mesogastrium?*)

Greater omentum: double layer of peritoneum extending from stomach to transverse colon (*mesogastrium?*)

Peritoneal ligaments: double layer of peritoneum that connects one organ to another organ or to the abdominal wall. May form part of an omentum. They support the organ in position and may convey their neurovascular structures.

Mesentery: Peritoneal folds that attach viscera to posterior abdominal wall

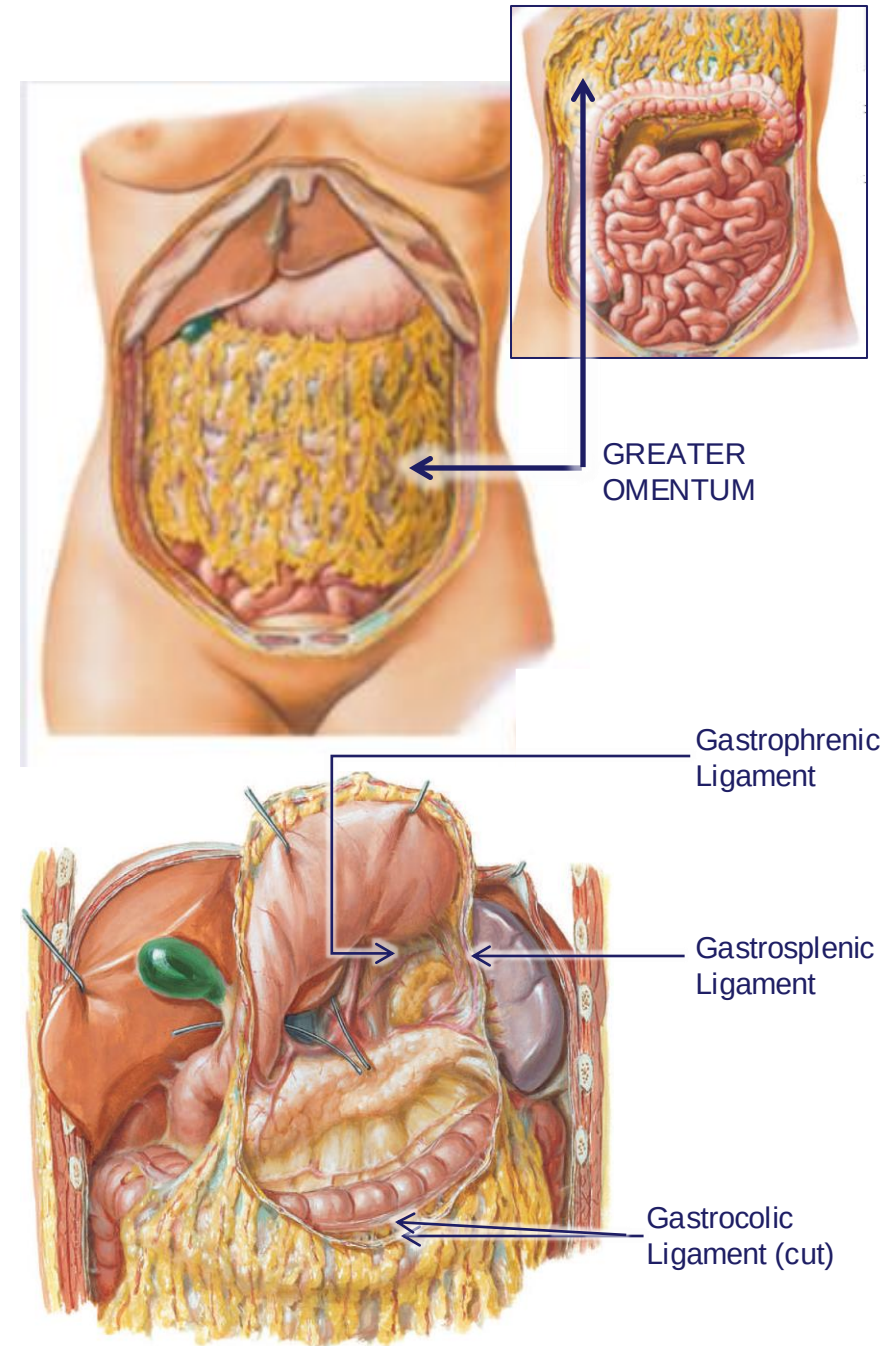


Greater Omentum

- Attaches greater curvature of stomach to transverse colon
- Has descending and ascending portions that usually fuse together forming a four layered (2 anterior+2 posterior) fatty "omental apron"
- Normally referred to as the policeman of the abdomen
- Contain gastro-omental vessels

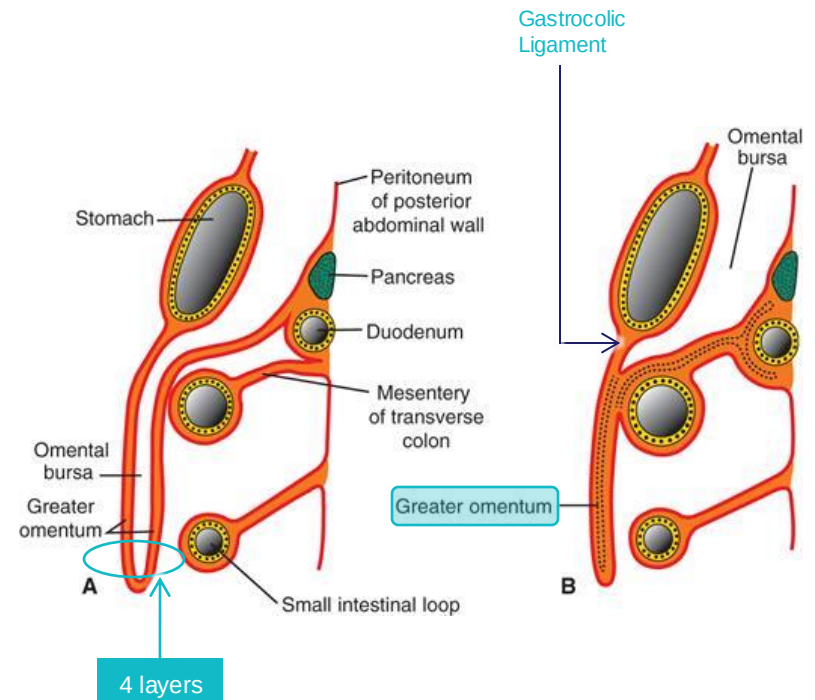
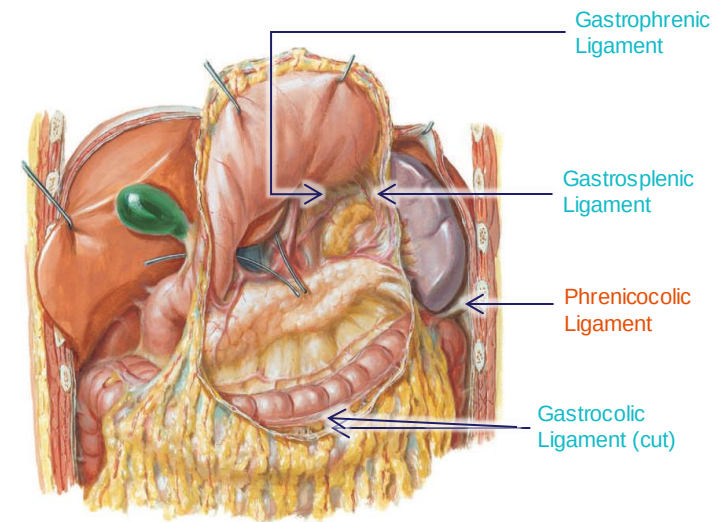
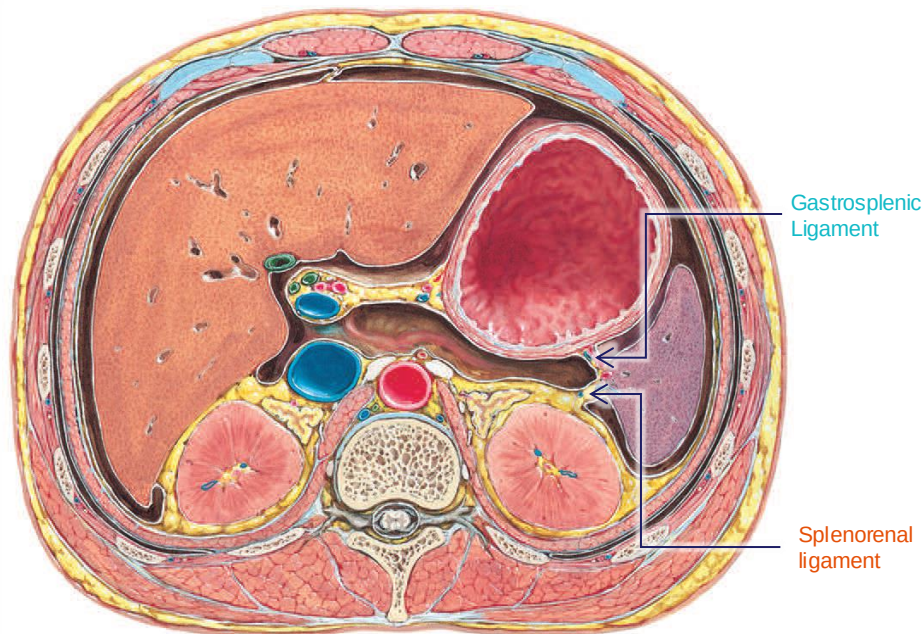
Ligaments

- **Gastrocolic** ligament – stomach to the transverse colon and mesocolon
- **Gastrosplenic** ligament – stomach to spleen
- **Gastrophrenic** ligament – stomach to diaphragm



Greater Omentum

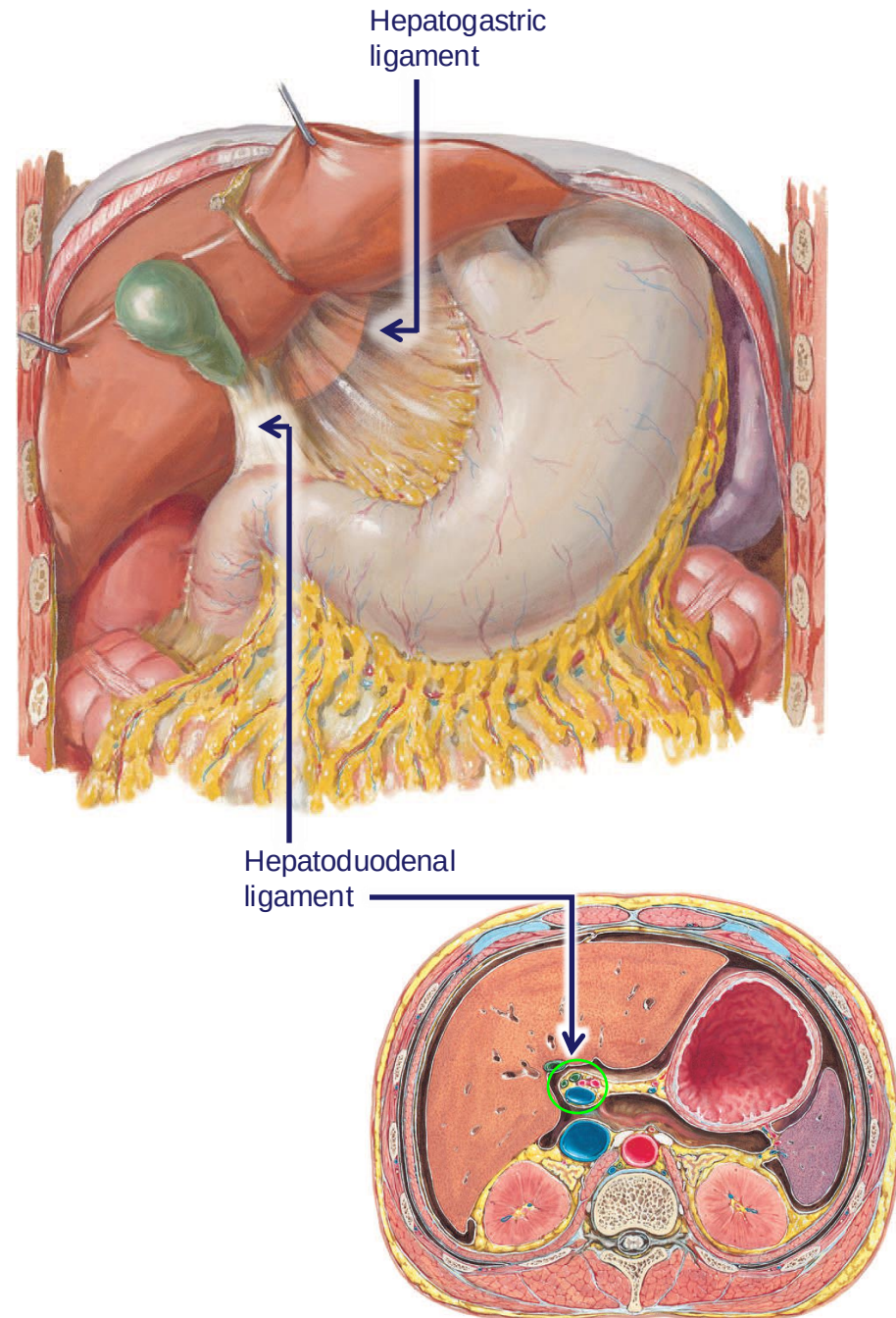
- **Gastrophrenic** ligament – stomach to diaphragm
- **Gastrosplenic** ligament – stomach to spleen
- **Gastrocolic** ligament – stomach to the transverse colon and mesocolon
- **Splenorenal** ligament – spleen to left kidney
- **Phrenicocolic** ligament – colon to diaphragm



Lesser Omentum

Extends from the lesser curvature of the stomach and proximal duodenum to the inferior surface of the liver
Contains two ligaments:

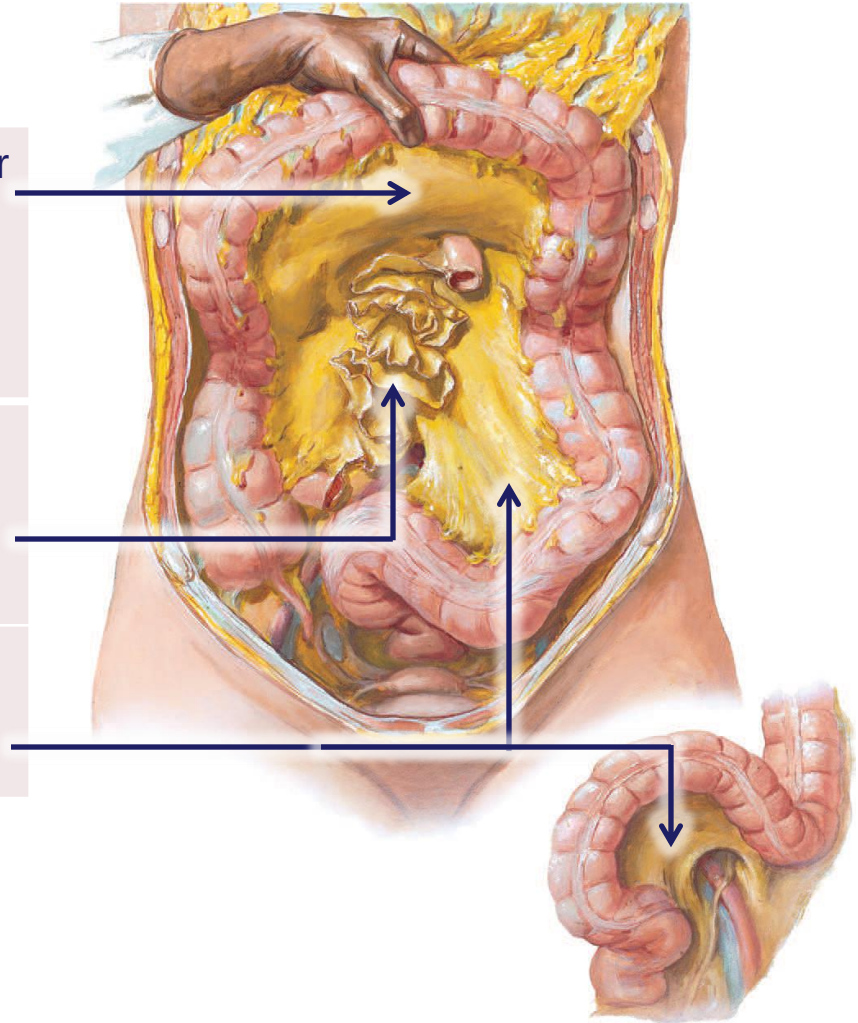
- Hepatogastric ligament – membranous portion
 - Contains: gastric arteries
- Hepatoduodenal ligament – thickened portion
 - Contains:
 1. Portal vein
 2. Proper hepatic artery
 3. Common bile duct



Mesenteries

- Peritoneal folds that attach viscera to posterior abdominal wall

Transverse mesocolon	Associated with transverse colon	Anterior layer adherent to posterior greater omentum
Mesentery (Proper)	Associated with parts of jejunum and ileum	Fan-shaped, oblique course
Sigmoid mesocolon	Associated with sigmoid colon	Inverted V-shaped fold



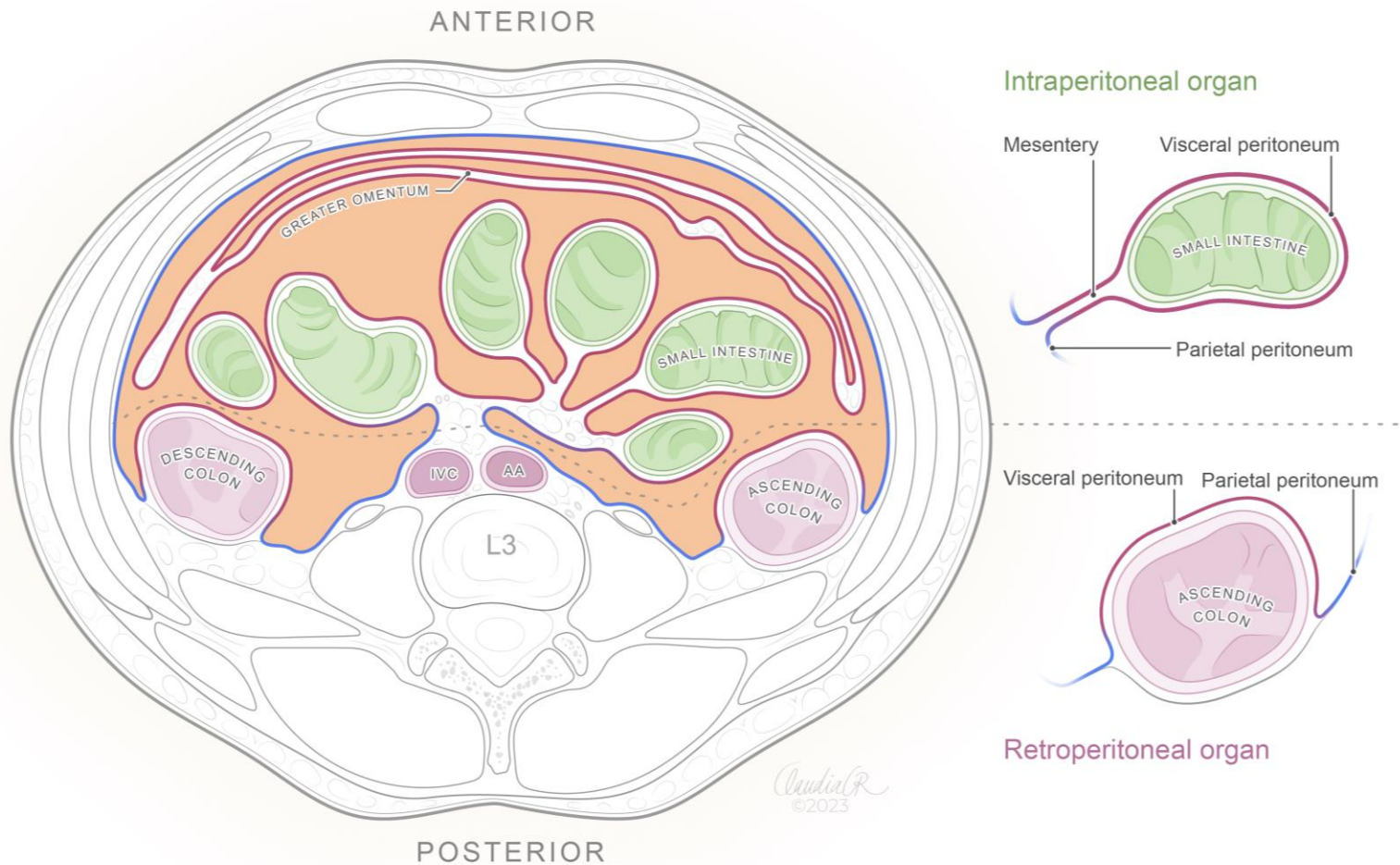
Peritoneal relationships of abdominal

Intraperitoneal Organs

Suspended in the cavity

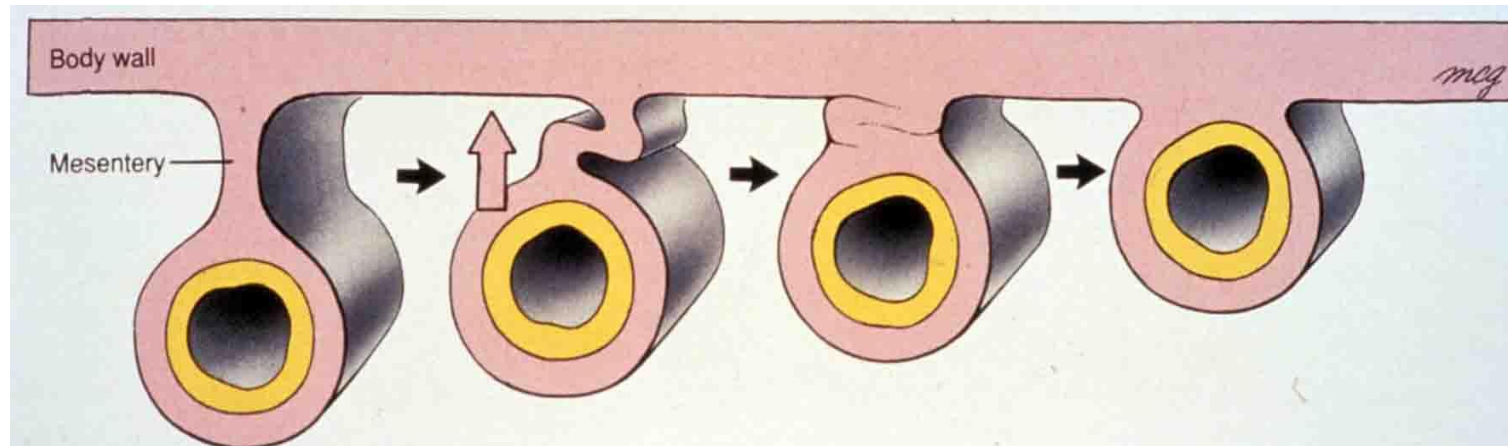
Retroperitoneal organs

Outside the peritoneal cavity



Peritoneal relationships of abdominal organs

- All gut structures start their development on a mesentery
- Throughout the course of development i.e., rotation of the gut, the **mesentery fuses with the posterior abdominal wall**. Peritoneum only covers the anterior surface. Therefore, the structure that was once intraperitoneal now is **secondarily retroperitoneal**.
- Secondary retroperitoneal structures are fixed (immobile)



Peritoneal relationships of abdominal organs

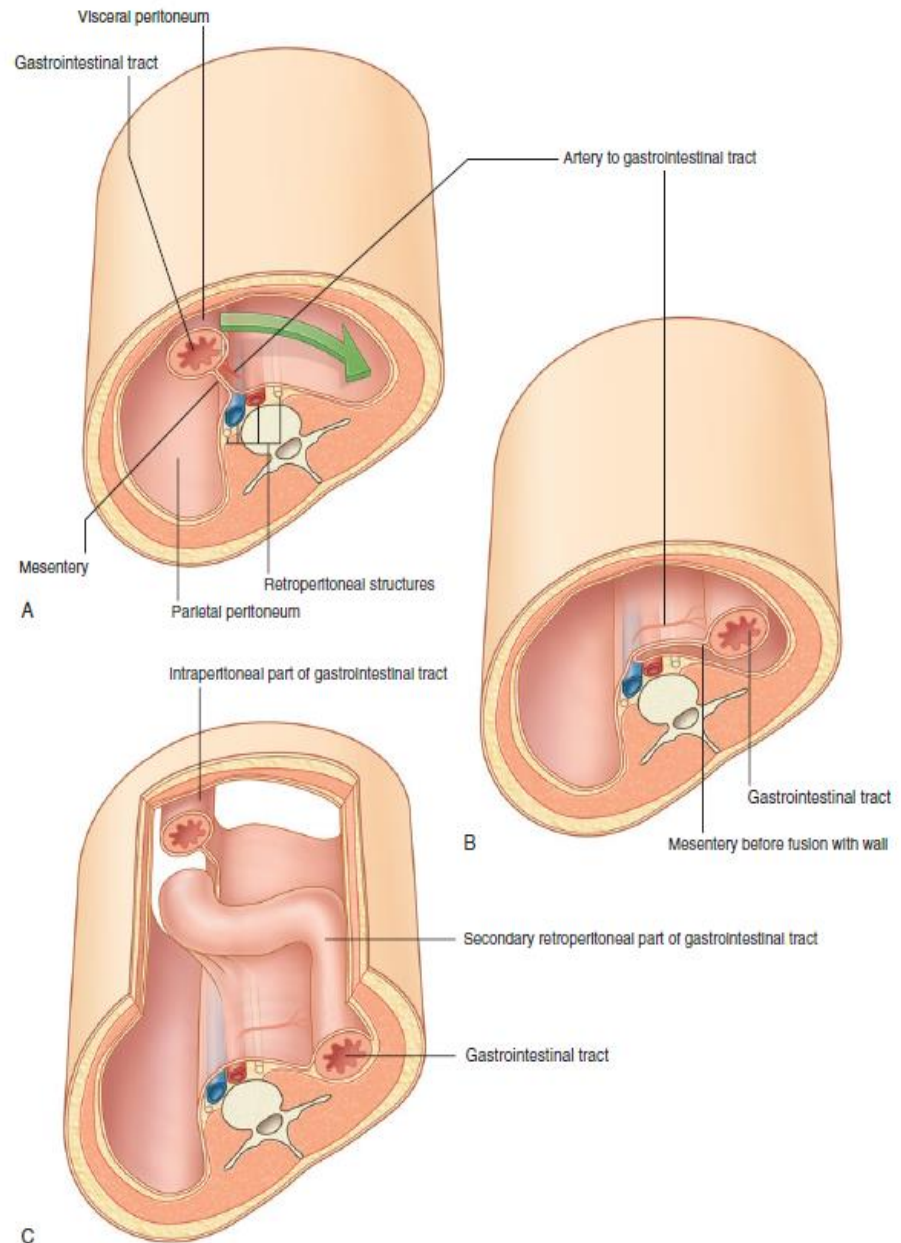


Fig. 4.7 A series showing the progression (A to C) from an intraperitoneal organ to a secondarily retroperitoneal organ.

Peritoneal relationships of abdominal organs



Memorize

Intraperitoneal organs <i>(have a mesentery)</i>	Primarily Retroperitoneal <i>(posterior body wall behind peritoneum)</i>	Secondarily Retroperitoneal
• Stomach • Liver • Duodenum (1st part) • Small intestine (jejunum, ileum) • Spleen • Cecum and Appendix • Transverse & Sigmoid colon	• Kidneys • Ureters • Aorta • IVC • Sympathetic trunks	• Pancreas • Duodenum 2nd, 3rd & 4th part • Ascending colon • Descending colon • Rectum

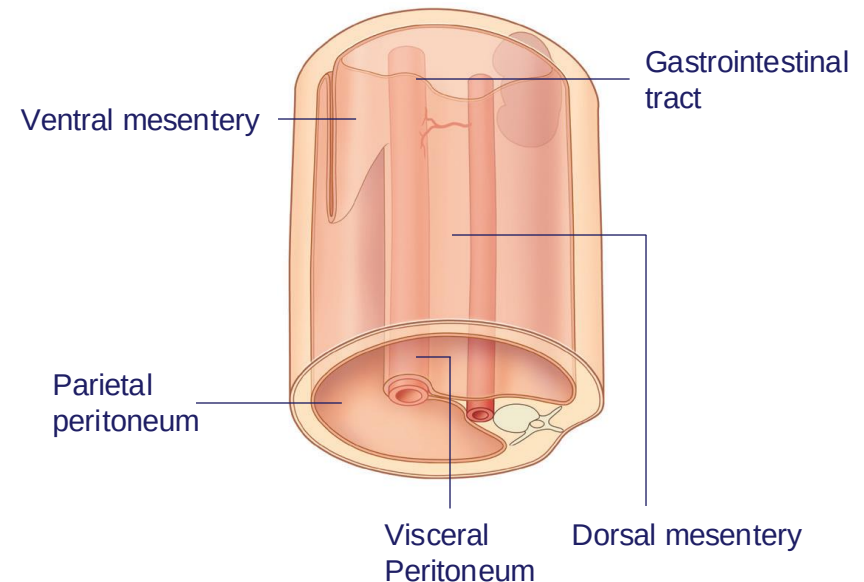
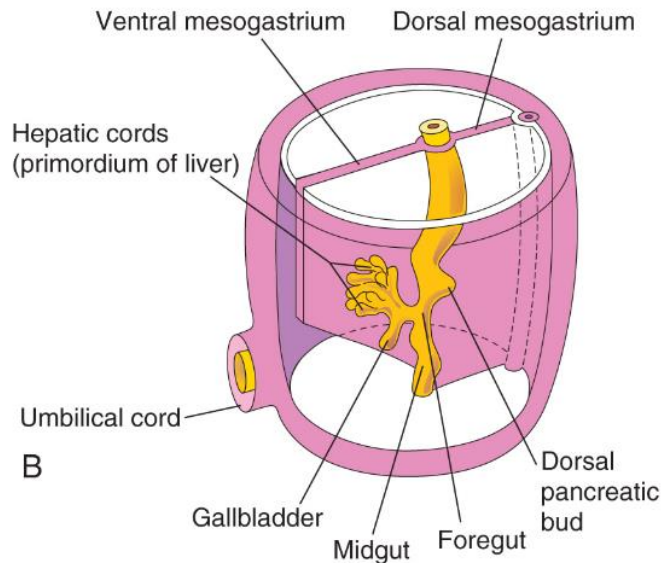
Peritoneum – Embryologic origin

Ventral Mesentery

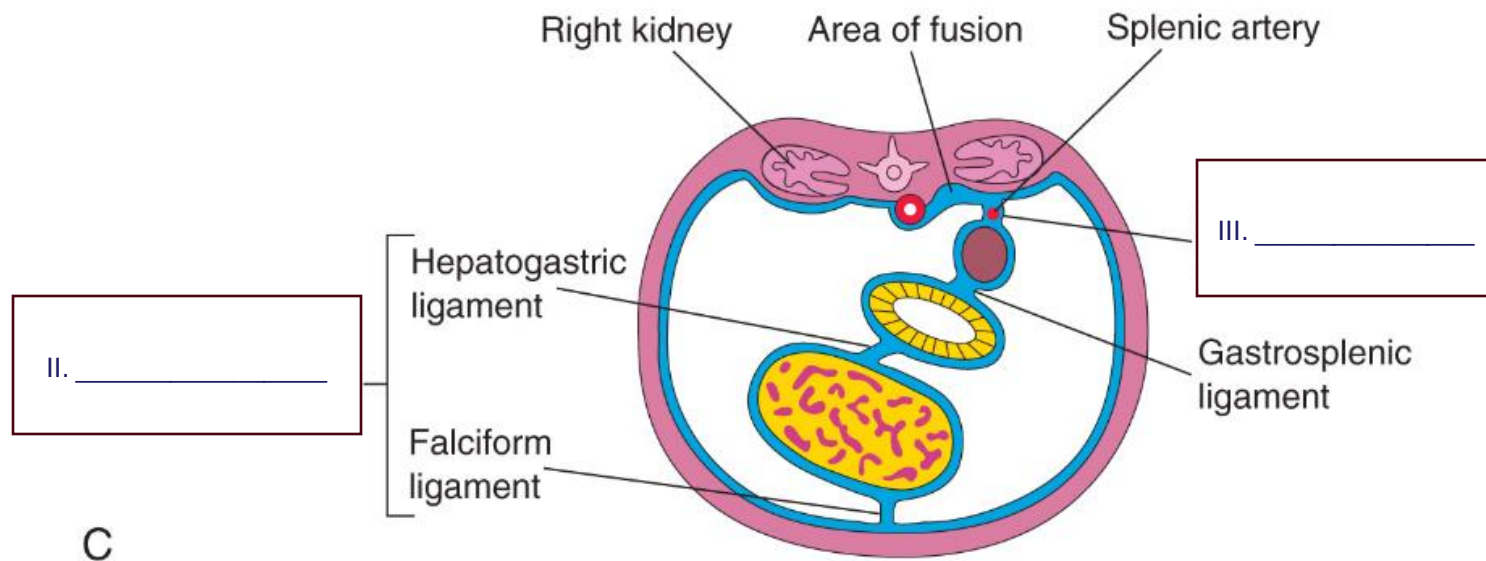
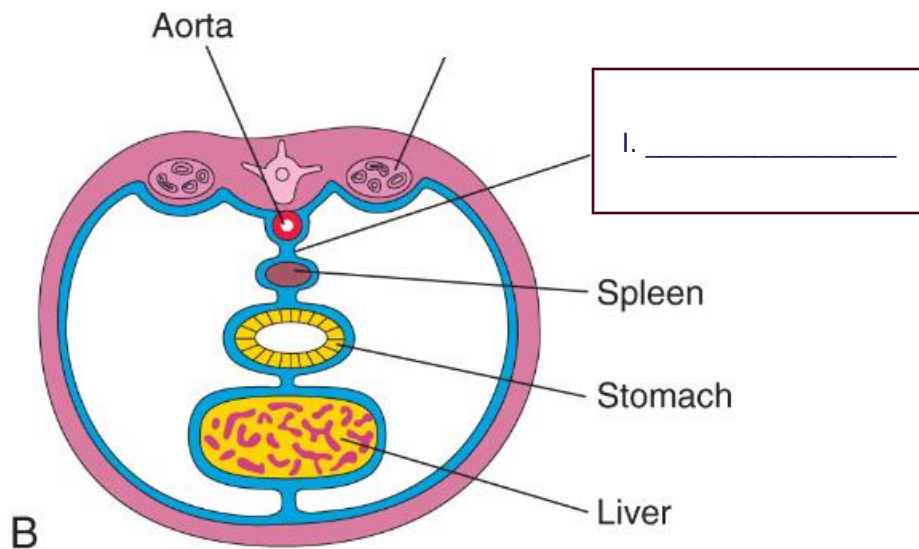
- From ventral mesogastrium
- Attaches stomach & duodenum (foregut) to the liver & ventral abdominal wall
- Falciform ligament
- Lesser omentum
 - Hepatogastric ligament
 - Hepatoduodenal ligament

Dorsal Mesentery

- From dorsal mesogastrium
- Suspend foregut, midgut and hindgut from dorsal abdominal wall
- Greater omentum
- Gastrosplenic ligament
- Splenorenal ligament



Fill in the Blanks



Subdivisions of Peritoneal Cavity

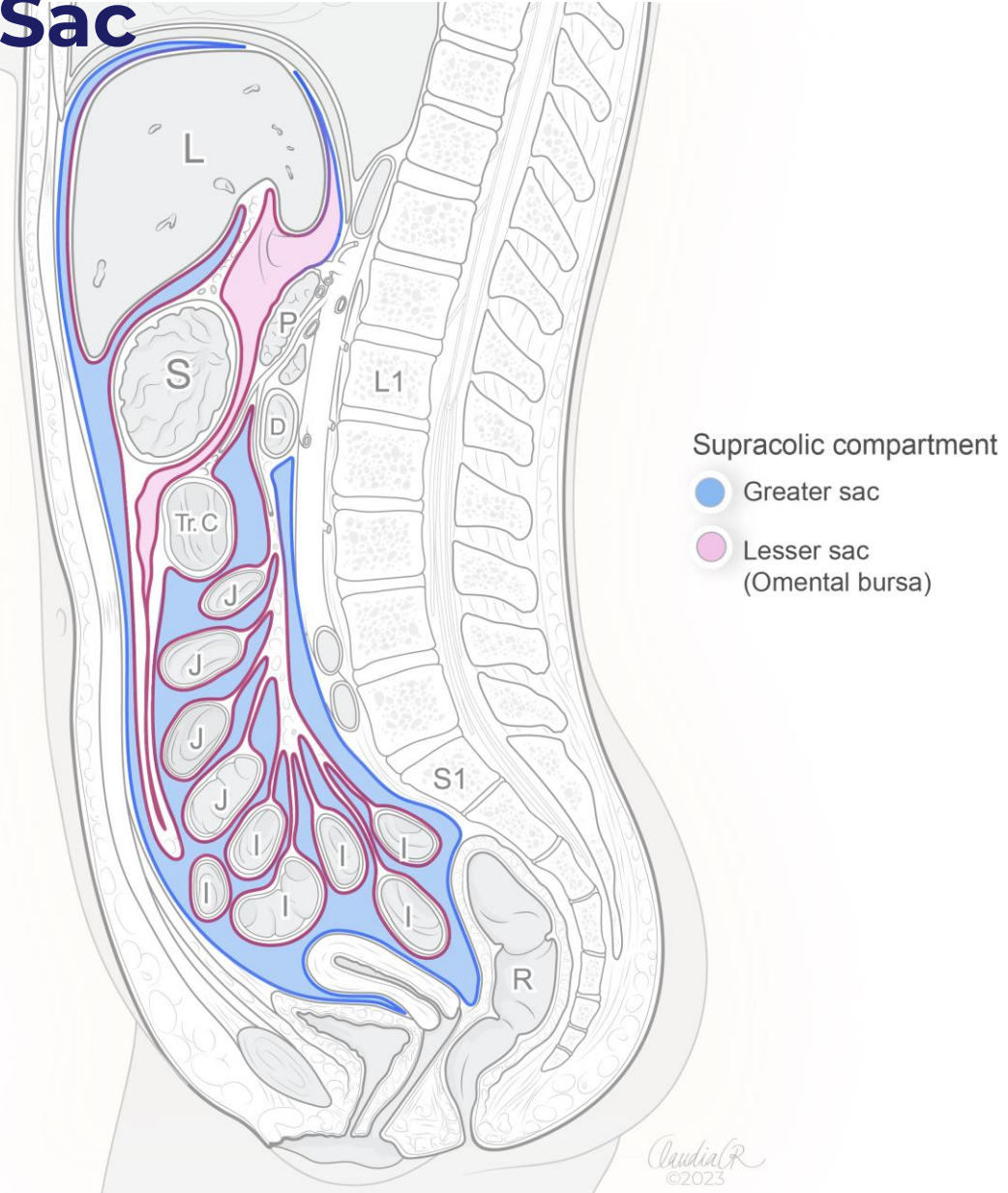
Greater & Lesser Sac

Greater Sac

- Main and larger part of the peritoneal cavity.
- Superior – diaphragm
- Inferior – pelvic cavity
- Anterior & Posterior – Abdominal walls

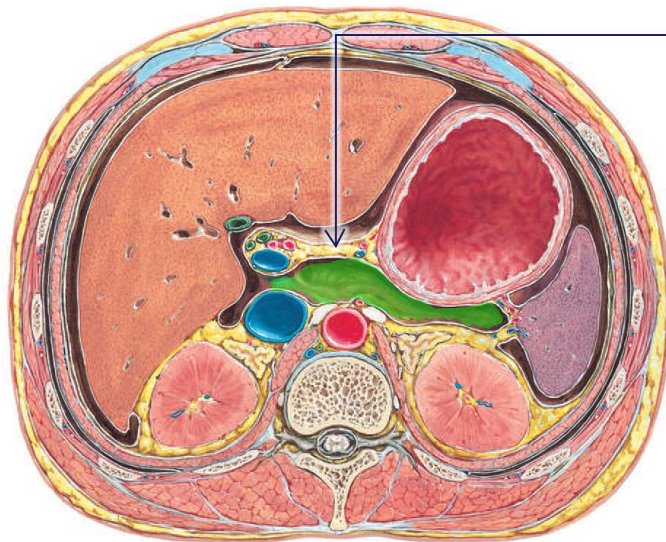
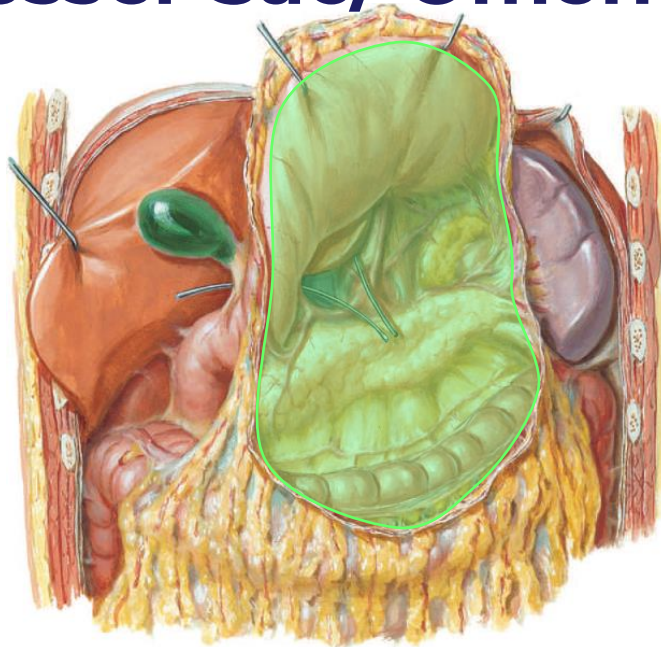
Lesser Sac

- Smaller part of the peritoneal cavity, lies posterior to the stomach, lesser omentum and adjacent structures

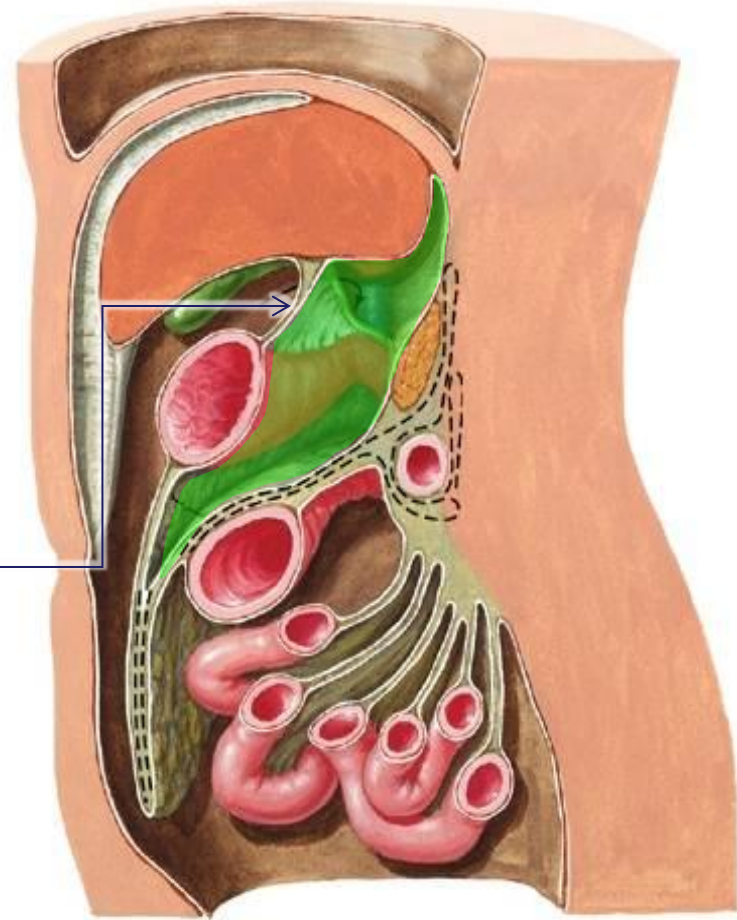


Subdivisions of Peritoneal Cavity

Lesser Sac/ Omental Bursa

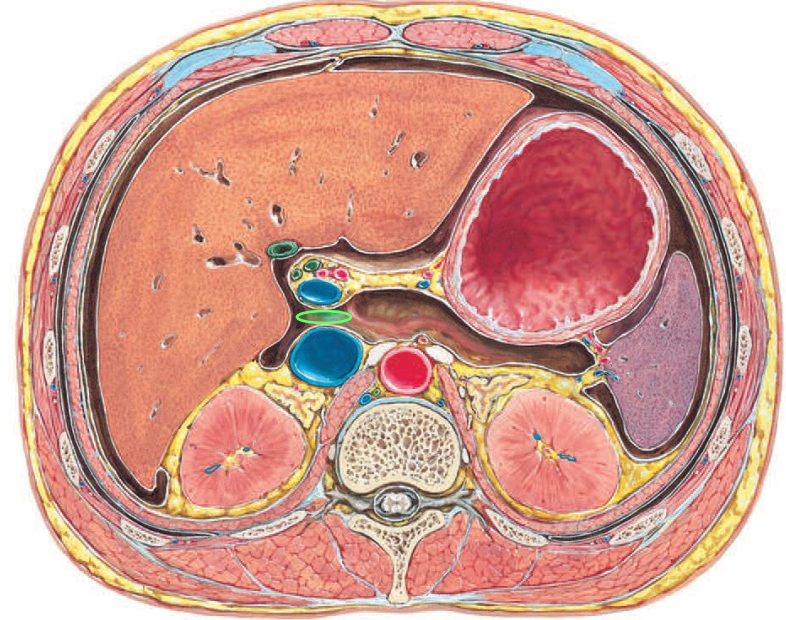
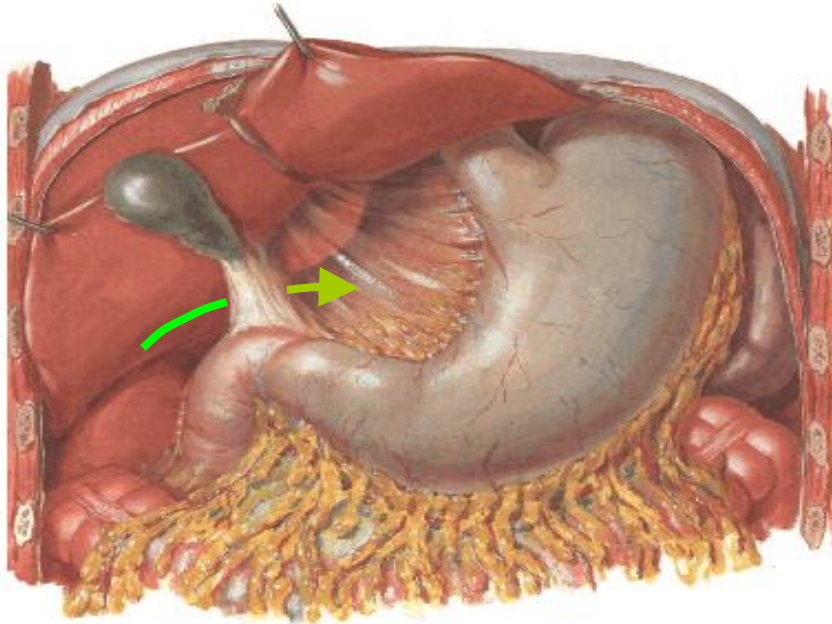


Lesser
omentum



Subdivisions of Peritoneal Cavity

Omental (Epiploic) Foramen (of Winslow)



Location	Anatomical structure
Anterior	Hepatoduodenal ligament with portal triad
Inferior	Duodenum (Superior part)
Posterior	Inferior vena cava
Superior	Caudate lobe of liver

Allow for communication between greater and lesser sac

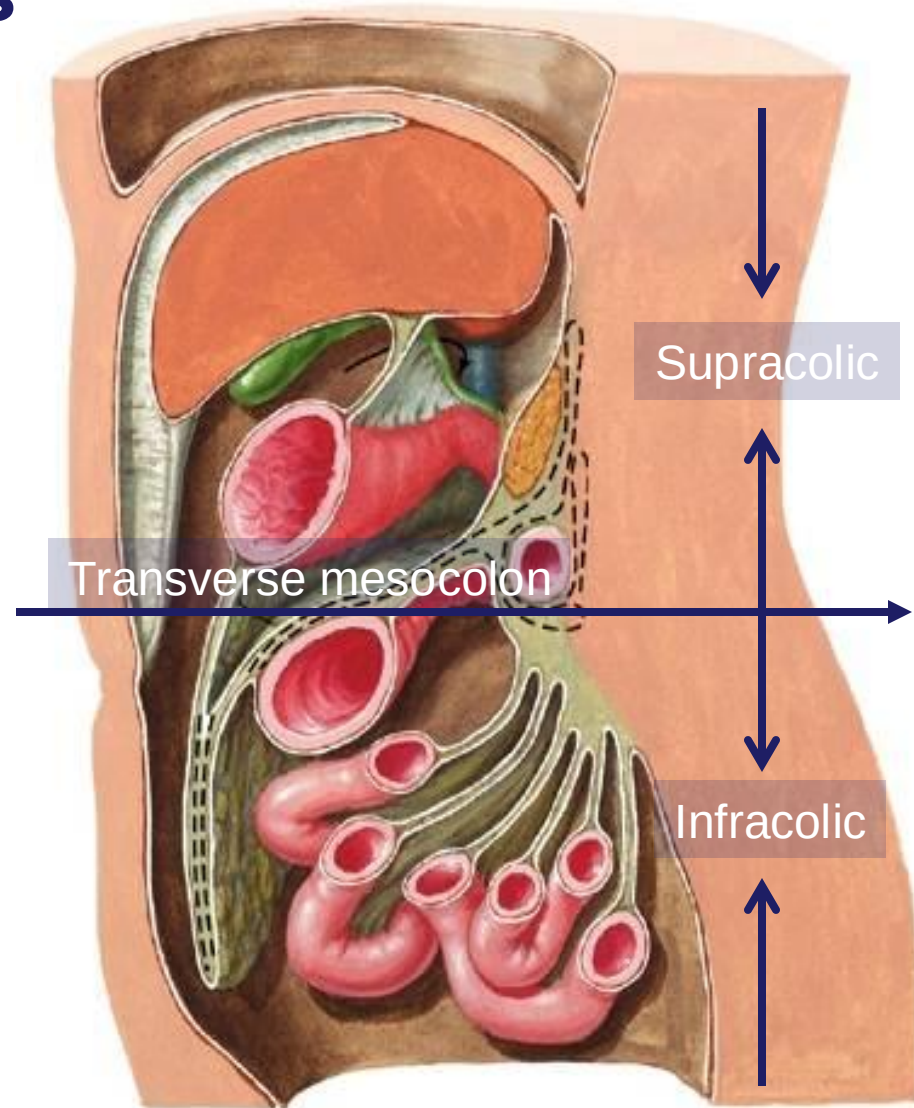
Compression of the hepatoduodenal ligament (Pringle maneuver) prevents inflow of blood into the liver thereby reducing bleeding in the liver as a result of traumatic injury

Subdivisions of Peritoneal Cavity

Colic Compartments

Mesentery of transverse colon divides peritoneal cavity into 2 compartments

- **Supracolic compartment** contains:
 - Stomach
 - Liver
 - Spleen
- **Infracolic compartment** contains:
 - Small intestine
 - Ascending & descending colon
- Lies posterior to greater omentum
- Further divided into right and left infra colic spaces by the small intestine.



Hepatorenal Recess (Pouch of Morrison)

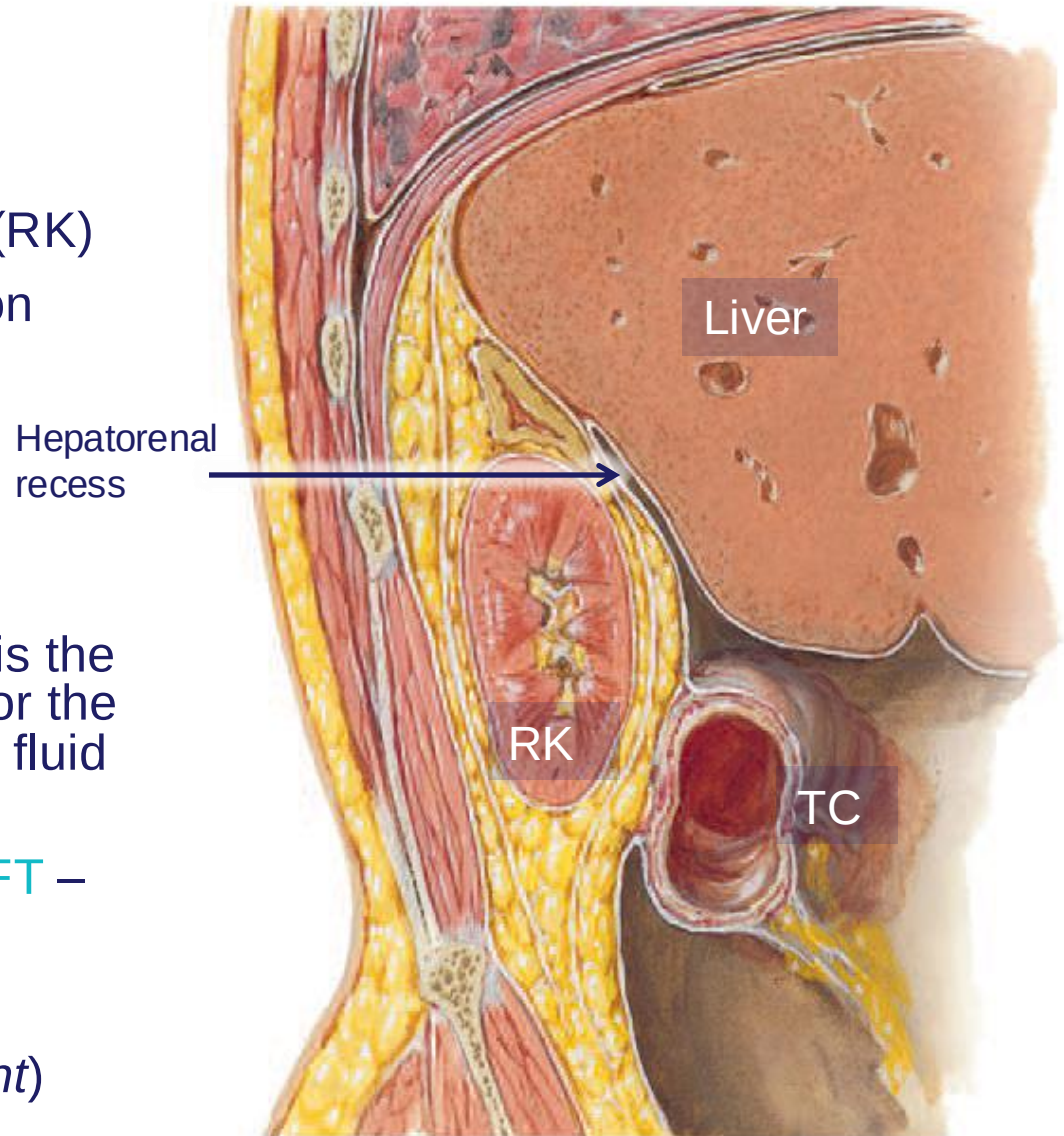
Boundaries

- **Anterior:** Liver
- **Posterior:** Right kidney (RK)
- **Inferior:** Transverse colon (TC)

Clinical correlate:

- The hepatorenal recess is the most dependent space for the collection of pathological fluid in the supine patient
- Similar space on the **LEFT** – **splenorenal recess**

(in Supracolic Compartment)

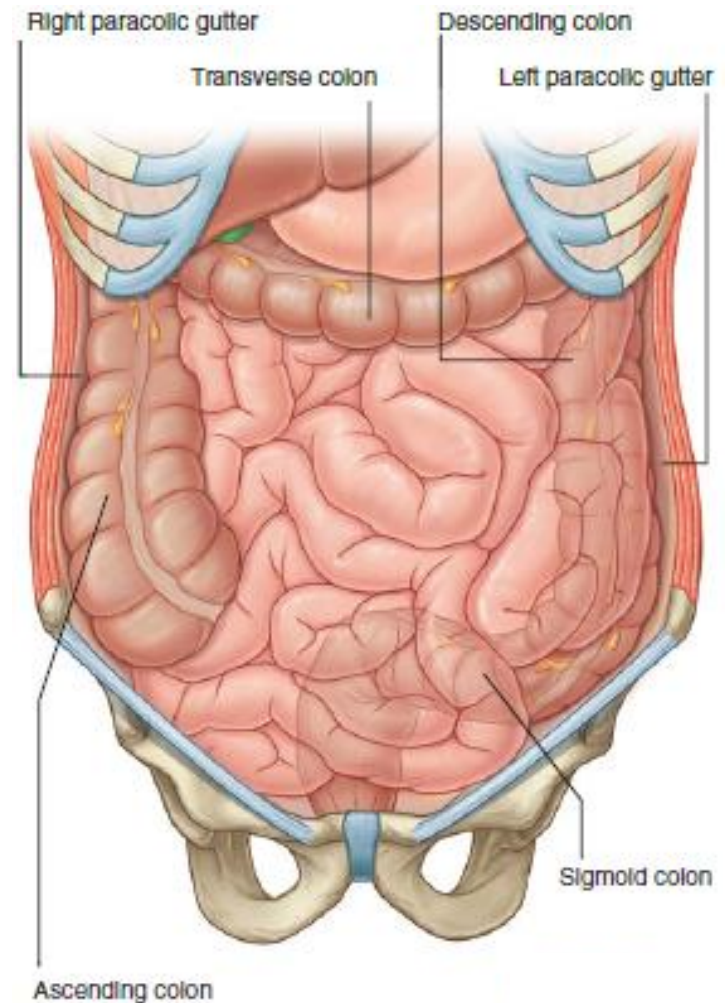


Subdivisions of Peritoneal Cavity

Paracolic Gutters

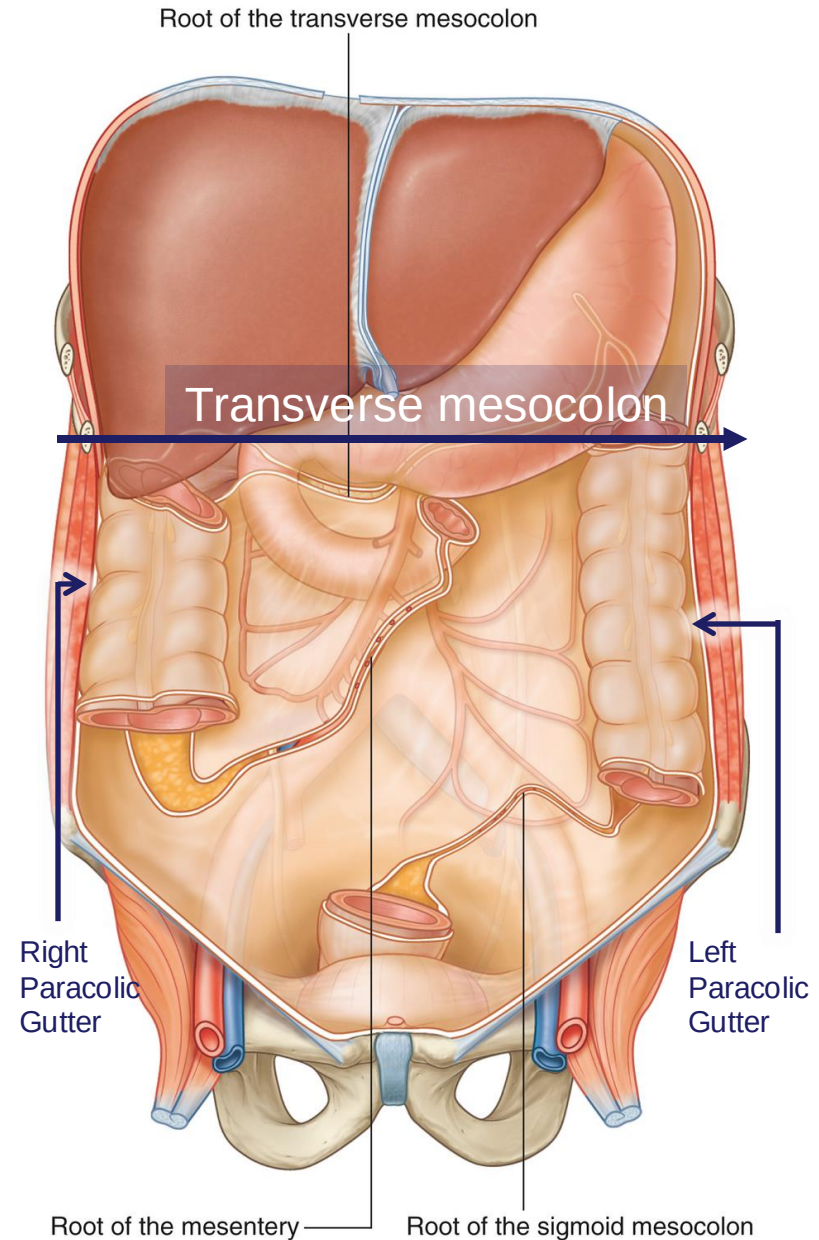
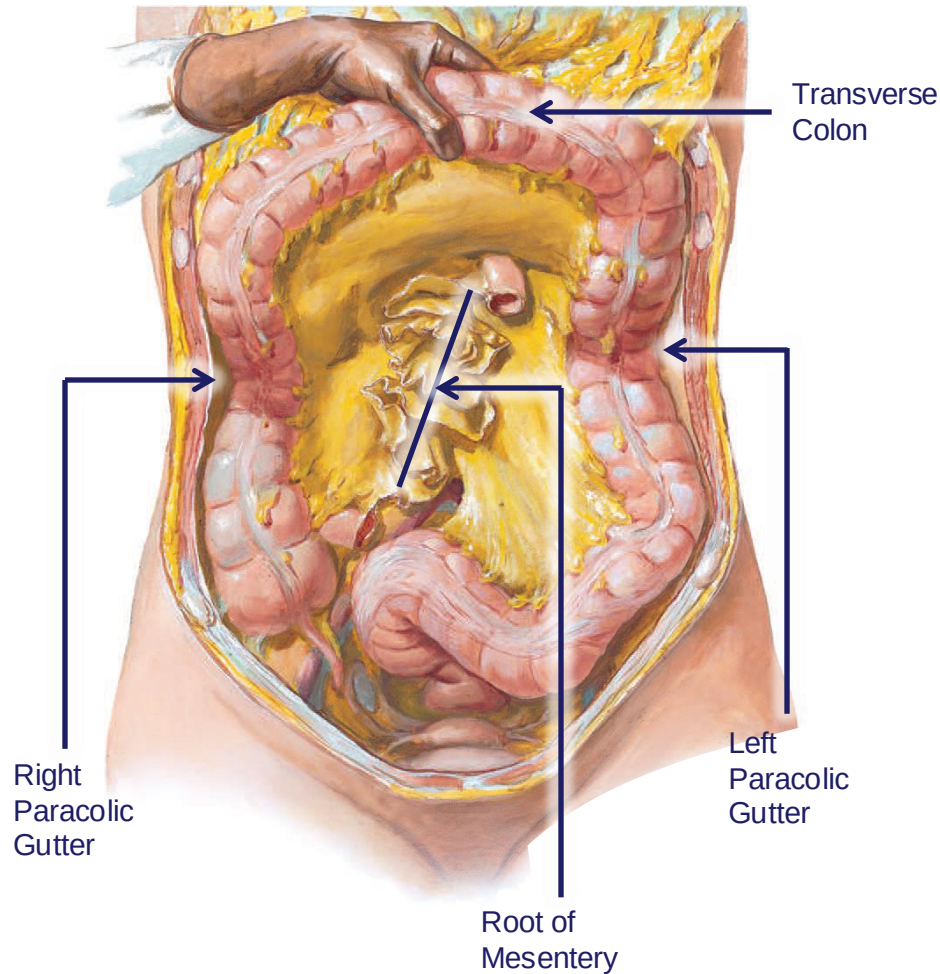
- Depressions between the lateral margins of the ascending and descending colon and the posterolateral abdominal wall.
- Allow for communication between the supracolic and infracolic compartments
- Allow for relatively blood-free mobilization of the ascending and descending colon by cutting the peritoneum along these lateral paracolic gutters
 - Major vessels and lymphatics on medial or posteromedial sides of ascending and descending colon
- The left paracolic gutter is narrower and partially obstructed by the phrenicocolic ligament.
 - Easier for infection to travel to the right subphrenic space

(in Infracolic Compartment)

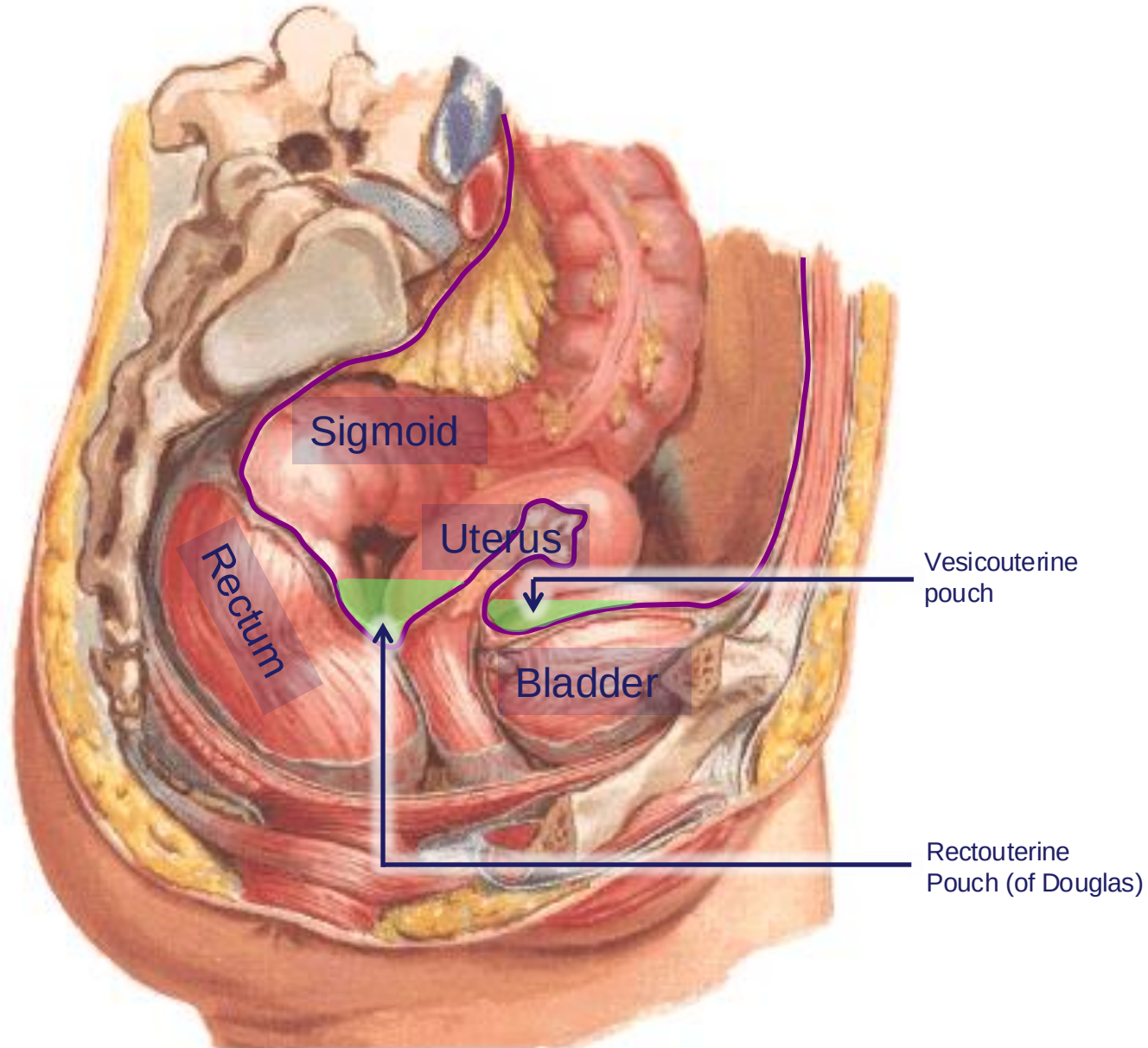


Subdivisions of Peritoneal Cavity

Paracolic Gutters



Peritoneal spaces - Pelvis

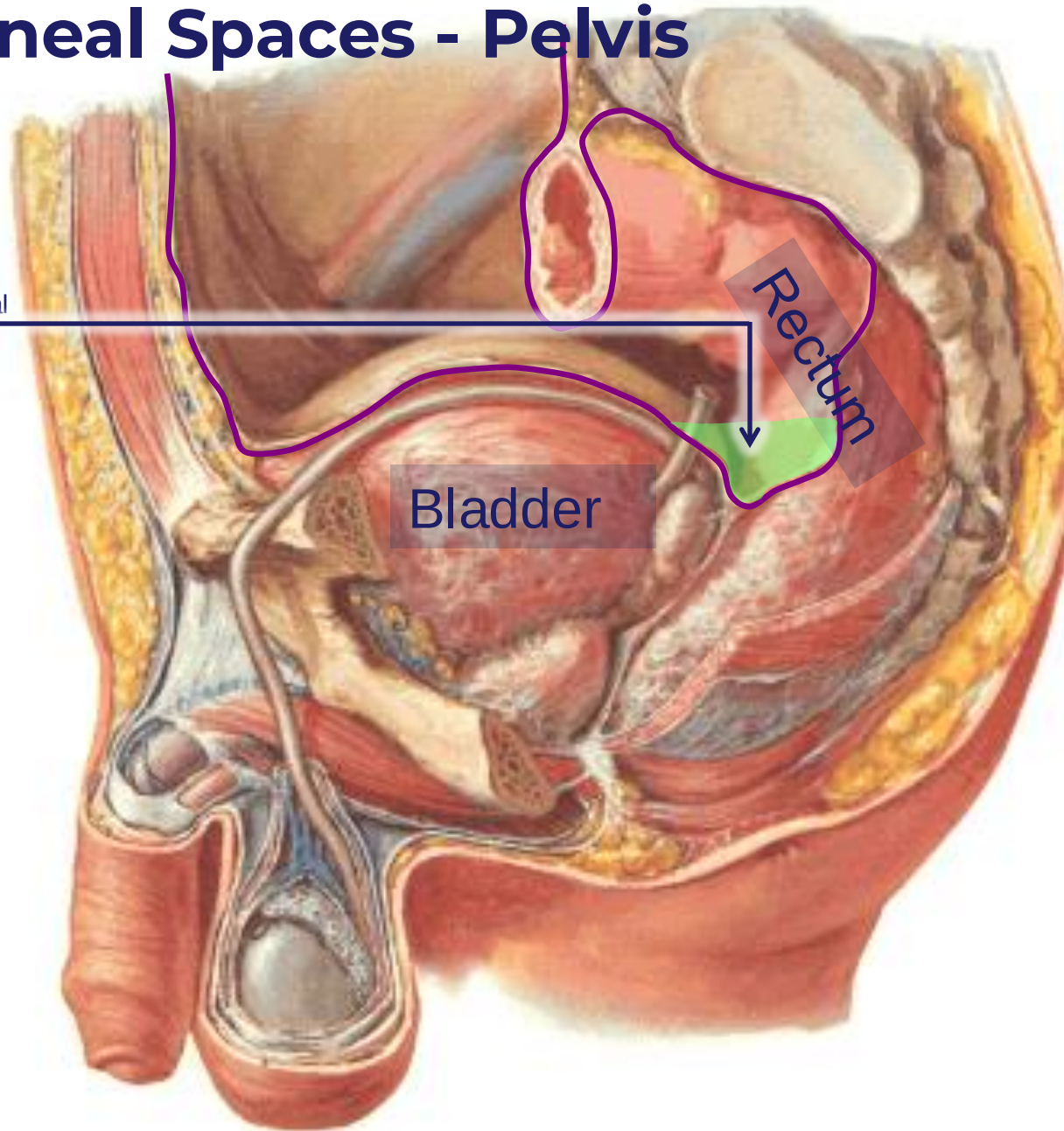


Peritoneal Spaces - Pelvis

Recto-vesical
pouch

Rectum

Bladder



Direction of flow of the peritoneal fluid

